**Algorithm 1:** $\theta_{k+1} \leftarrow \texttt{DistributedFixedRolloutUpdate}(\theta_k, (\texttt{TrainingStorage}_g)_g, \mathsf{m})$

**Parameters:** immutable GPU-local stores produced by the matching target builder; opaque $(\texttt{PartitionSpec}_m)_{m\in\mathcal{M}}$; $\mathsf{m} \in \{\mathsf{normal}, \mathsf{seqRB}, \mathsf{branchRB}\}$

**Parameters:** the production-v1 loss family: clipped PPO policy surrogate, value regression with optional value clipping, entropy, and legal-action KL to an optional frozen reference $\pi^{\theta_{\mathrm{ref(k)}}}$; synchronous optimizer, AMP rule, global clipping rule, and $K$ optimizer steps. $Q$-losses, RNaD/NeuRD, $W/S$ auxiliaries, and adaptive branching intensity are outside v1

**Hyperparameters:** per-partition measured statistical-batch time curves $C_{g,m}(B;\mathcal{C})$; finite calibrated physical shape sets $\mathcal{B}_{g,m}$; hard training and reference-build transient-byte predictors $W_{g,m}(J;\mathcal{C})$ and $W^{\mathrm{refbuild}}_{g,m}(J;\mathcal{C})$; time budgets $T_g$; lagged cheap variance proxies $\widehat{\sigma}^2_{g,m}$; usable bytes $M_{\mathrm{device}}$ and unused guard $M_{\mathrm{guard}}$, with $M_{\mathrm{hard}} = M_{\mathrm{device}} - M_{\mathrm{guard}}$

**Definitions:** $\mathcal{D}_{g,m} = \{0, \dots, D_{g,m} - 1\}$ and

$$\log \omega_{g,m,d} = \begin{cases} 0, & \mathsf{m} \neq \mathsf{branchRB}, \\ -\sum_{s \prec (m,d)} \log N(s), & \mathsf{m} = \mathsf{branchRB}; \end{cases} \qquad \log Z_{g,m} = \mathrm{logsumexp}_{d\in\mathcal{D}_{g,m}} \log \omega_{g,m,d}.$$

The target finalizer stores $\log Z_{g,m}$ and the normalized alias law below, then releases the temporary $\log \omega$ rows

**Definitions:** For $\log Z_{g,m} > -\infty$, $q_{g,m}(d) = e^{\log \omega_{g,m,d} - \log Z_{g,m}}$ and $\alpha_{g,m} = e^{\log Z_{g,m} - \log Z}$, where $\log Z = \mathrm{logsumexp}_{g,m} \log Z_{g,m}$

**Definitions:** $\mathcal{X}_m$ is the exact loss domain, $\mathcal{Y}_m$ the natural model-row domain, and $\varphi_m : \mathcal{X}_m \to \mathcal{Y}_m$ the fixed architecture lookup. Current, old, and reference quantities at exact state $x$ are read directly from model row $\varphi_m(x)$

**Definitions:** $M^{(0)}_g$ is resident optimization storage excluding optional reference caches and one physical-subspan transient, but including immutable training storage, model/reference/optimizer state, the flat gradient buffer, collective workspace, and one preallocated int32 index buffer sized to the largest statistical count allowed by $T_g$. For cache set $\mathcal{C}$ define

$$M^{\mathrm{base}}_g(\mathcal{C}) = M^{(0)}_g + \sum_{m\in\mathcal{C}} R_{g,m}, \qquad A_g(\mathcal{C}) = M_{\mathrm{hard}} - M^{\mathrm{base}}_g(\mathcal{C}).$$

$W_{g,m}(J;\mathcal{C})$ bounds only the simultaneously live input/target gathers, branched action spans, cached-output gathers or uncached reference forward, current forward/backward activations, exact-loss reductions, AMP temporaries, and backend workspace for decision-index subspan $J$

**Definitions:** $W^{\mathrm{refbuild}}_{g,m}(J;\mathcal{C})$ bounds the simultaneously live input gather, frozen-reference forward/output, cache scatter, and backend workspace while building partition $m$. Resident bytes for the entire final cache set are already charged in $M^{\mathrm{base}}_g(\mathcal{C})$, making the admission test conservative during partial construction; the maximum over an empty cache set is defined as zero

**Definitions:** $\texttt{PhysicalPlan}_{g,m}(I, A; \mathcal{C})$ preserves the order of index vector $I$ and returns subspans $J$ using a small warmed set $\mathcal{B}_{g,m}$ plus an exact tail. Every $J$ satisfies $W_{g,m}(J;\mathcal{C}) \le A$; fail if a singleton cannot fit. One reusable staging arena serves every partition, and calibrated $W$ uses allocator-reserved high-water bytes, not merely live tensor bytes

**Definitions:** Activation checkpointing is off by default. It may be a separately calibrated member of $\mathcal{B}_{g,m}$ only if its recomputation is included in $C_{g,m}$ and it improves measured end-to-end spots/second, or if it is required to admit even one decision under the hard byte bound

**Definitions:** $\widehat{\sigma}_{g,m} = 1$ until observations exist. Any update is a lagged EMA of cheap scalar loss/advantage/residual statistics already produced by physical subspans; it never computes per-example parameter gradients or copies a full gradient buffer

**Output:** identical $\theta_{k+1}$ and optimizer state on every GPU

**foreach** $g \in \mathcal{G}$ ***as soon as its target builder finishes*** **do**

Read $(\log Z_{g,m}, \texttt{AliasTable}_{g,m})_m$; verify their decision domains and launch $\mathcal{Z}^\star \leftarrow \texttt{IAllGather}((\log Z_{g,m})_m)$ in the common collective order without an explicit preceding barrier; Assert $\texttt{TrainingStorage}_g$ contains no topology, leaf reach, completion result, frontier, rollout queue/state, or mutable occurrence;

**end**

**foreach** $g \in \mathcal{G}$ ***in parallel while the reduction progresses*** **do**

First prove that the uncached singleton path fits; otherwise checkpoint and stop cleanly without resampling the rollout. For each $m$, measure retained reference-output bytes $R_{g,m}$ and predict total learner time, including cache construction and any physical-plan changes. Select a subset

$$\mathcal{C}_g \in \arg\min_{C\subseteq\mathcal{M}} \widehat{C}^{\mathrm{total}}_g(C) \quad \text{s.t.} \quad M^{\mathrm{base}}_g(C) + \max\left\{ \max_{m,d:\,\log Z_{g,m} > -\infty} W_{g,m}(\{d\}; C),\ \max_{m\in C, J} W^{\mathrm{refbuild}}_{g,m}(J; C) \right\} \le M_{\mathrm{hard}},$$

where the finite search always includes $C = \varnothing$. Transactionally reserve all cache destinations before the first build forward. For $m \in \mathcal{C}_g$, evaluate the frozen reference once on every distinct retained input using bounded partition-$m$ physical subspans, scatter its natural $\mathcal{Y}_m$-row outputs, and release each build subspan immediately. If reservation or a calibrated build contract fails, release every partial cache and use $C = \varnothing$ without changing targets or random draws. For $m \notin \mathcal{C}_g$, recompute the reference only for the sampled subspan;

**end**

$\mathcal{Z} \leftarrow \texttt{Await}(\mathcal{Z}^\star)$; compute $\log Z$ by log-sum-exp and assert $\log Z > -\infty$;

Choose arbitrary integers $B_{g,m} \ge 1$ for every positive-mass stratum by minimizing

$$\sum_{m:\log Z_{g,m} > -\infty} \frac{\alpha^2_{g,m} \widehat{\sigma}^2_{g,m}}{B_{g,m}} \quad \text{s.t.} \quad \sum_m C_{g,m}(B_{g,m}; \mathcal{C}_g) \le T_g,$$

using $B_{g,m} \propto \alpha_{g,m}\widehat{\sigma}_{g,m}/\sqrt{c_{g,m}}$ only as an integer-search warm start. $B_{g,m}$ is a statistical sample count: it is not rounded to 32 or to any physical shape quantum; $C_{g,m}$ includes the measured subspan plan, tail, and launch overhead; On every GPU allocate one fp32 flat SUM buffer $G_g$ in fixed parameter order; bind every parameter's gradient directly to its corresponding view in $G_g$ `/* no per-step flatten/copy buffer */`

$\theta \leftarrow \theta_k$;

**for** $t = 0, \dots, K-1$ **do**

**foreach** $g \in \mathcal{G}$ ***in parallel*** **do**

Zero $G_g$ once; **foreach** $m : \log Z_{g,m} > -\infty$ **do**

Draw only the integer index vector $I_{g,m} = (D_{g,m,b})_{b=1}^{B_{g,m}} \overset{\mathrm{iid}}{\sim} q_{g,m}$ from its fixed alias table (implicit uniform in sequential modes); $\mathcal{J}_{g,m} \leftarrow \texttt{PhysicalPlan}_{g,m}(I_{g,m}, A_g(\mathcal{C}_g); \mathcal{C}_g)$; **foreach** $J \in \mathcal{J}_{g,m}$ **do**

Gather only $J$'s aligned partition inputs, decision targets/posterior references, branched action spans when present, and cached reference rows; if $m \notin \mathcal{C}_g$, evaluate the frozen reference on $J$ under no-grad and retain its output only through this loss; evaluate the current model and immediately backward, with no retained graph, the common-AMP-scale contribution

$$\frac{\alpha_{g,m}}{B_{g,m}} \sum_{d\in J} \texttt{ModeLoss}_{\mathsf{m}}(\theta; (m,d), \texttt{TrainingStorage}_g)$$

directly into the gradient views; update only cheap scalar proxy statistics; release the graph, all gathers, uncached reference output, loss reductions, and subspan staging before the next $J$;

**end**

Release $I_{g,m}$; never gather all $B_{g,m}$ rows, normalize a physical subspan, globally sort decisions, inspect the meaning of $m$, or permute exact private-state rows;

**end**

**end**

$G \leftarrow \texttt{AllReduceSum}((G_g)_g)$ with no division by $|\mathcal{G}|$; derive the identical nonfinite decision from $G$, unscale once, apply the common global clipping rule once, and perform the identical optimizer/loss-scale transition on every GPU;

**end**

$\theta_{k+1} \leftarrow \theta$; release reference caches and all training stores;

`/* No sampled loss is multiplied by ω again. For every current θ, conditionally on the fixed distributed rollout, */`

$$\mathbb{E}[G \mid (\texttt{TrainingStorage}_g)_g, \theta] = \nabla_\theta \sum_{g,m,d} e^{\log \omega_{g,m,d} - \log Z} \texttt{ModeLoss}_{\mathsf{m}}(\theta; (m,d), \texttt{TrainingStorage}_g).$$

**return** $\theta_{k+1}$;

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**Algorithm 2:** $\texttt{ModeLoss}_{\mathsf{m}}(\theta; (m,d), \texttt{TrainingStorage})$**: exact storage-to-math map**

Forward the sampled partition input under $\theta$ on $\mathcal{Y}_m$. For every exact loss state $x \in \mathcal{X}_m$, index current, old, and optional reference outputs at $\varphi_m(x)$; retain no duplicate $X_m$-wide model output; **if** $\mathsf{m} = \mathsf{normal}$ **then**

Set $\widehat{V} = \texttt{lambdaReturn}[d]$ and $\widehat{A} = \texttt{lambdaReturn}[d] - \texttt{oldV}[d]$ at the stored realized type/event; evaluate the production-v1 normal-mode loss;

**else if** $\mathsf{m} = \mathsf{seqRB}$ **then**

Read $w(-) = \texttt{SequentialPosteriorPool}[\texttt{posteriorRef}[d]]$ and $U(-) = \texttt{lambdaReturn}[d]$; set $V(-) = U(-)$ and $A(x) = U(x) - \texttt{oldV}[d, \varphi_m(x)]$; return the exact sum $\sum_{x\in X_m} w(x)\mathcal{L}_x$ of the sequential RB surrogate;

**else**

Let $[a_0, a_1)$ be $d$'s action span and $w_d(-) = (a_1 - a_0)^{-1} \sum_{a=a_0}^{a_1-1} \texttt{childPosterior}[a]$; evaluate the node value/entropy/reference terms with $w_d$ and $\texttt{valueHat}[d]$, plus

$$\frac{1}{a_1 - a_0} \sum_{a=a_0}^{a_1-1} \sum_{x\in X_m} \texttt{childPosterior}[a, x]\, \mathcal{L}^\pi_x(\texttt{advantageHat}[a, x]).$$

**end**

`/* PPO/value clipping, posterior sums, entropy, legal-action reference KL, and their production-v1 coefficients are exactly those in core_math.tex. The fixed φ_m lookup is part of the model architecture, not a storage transform. */`